

teradata.

# PROJECT SCION

Structural Change Intelligence Platform

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Product Tour

# What is SCION?

Proprietary structural intelligence platform for Teradata DNA

- Replaces Kalido — zero licensing cost, fully owned IP
- Monitors structural changes across the entire data warehouse
- Assesses blast radius and impact of every change
- AI-powered risk assessment and recommendations via TAISA
- 100% auditable — every decision backed by JSON evidence
- Governance scorecard, domain risk, volatility metrics

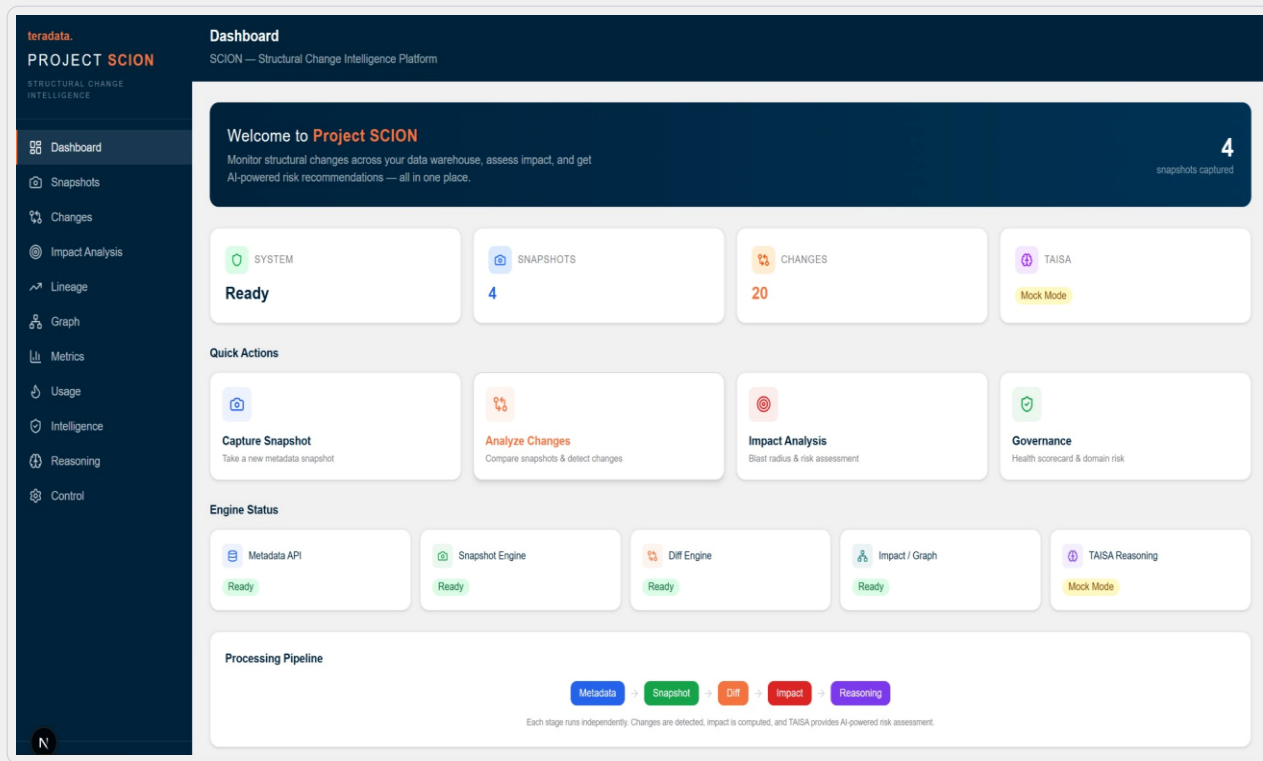
# 7 Engines, 1 Platform

Modular architecture — each engine operates independently

|   |                                |                                                        |
|---|--------------------------------|--------------------------------------------------------|
| 1 | <b>Metadata Ingestion</b>      | Multi-engine adapters, JSON import from parser         |
| 2 | <b>Snapshot Engine</b>         | State capture, structural hashing, versioning          |
| 3 | <b>Diff Engine</b>             | Severity scoring, breaking/non-breaking classification |
| 4 | <b>Graph &amp; Impact</b>      | Dependency graphs, blast radius, fragility metrics     |
| 5 | <b>Usage &amp; Criticality</b> | Query log ingestion, business criticality scoring      |
| 6 | <b>Intelligence Metrics</b>    | Governance scorecard, domain risk, volatility          |
| 7 | <b>TAISA AI Layer</b>          | LLM classification, recommendations, explanations      |

# Dashboard

Real-time platform status at a glance



## Key Features

- System status & engine health monitoring
- Quick actions for common workflows
- Processing pipeline visualization

# Snapshots — Metadata Capture

Version every state of your data warehouse

| ID | Created                 | Source       | Description                     | Active |
|----|-------------------------|--------------|---------------------------------|--------|
| 4  | 15/4/2026, 9:25:21 p.m. | api          | Snapshot created via API v1     |        |
| 2  | 15/4/2026, 8:56:37 p.m. | teradata_edw | Post-release structural changes |        |
| 1  | 8/4/2026, 8:56:37 p.m.  | teradata_edw | Weekly baseline - Banking EDW   |        |
| 3  | 16/3/2026, 8:56:37 p.m. | teradata_edw | Monthly baseline - Banking EDW  |        |

## Key Features

- Historical versioning with timestamps
- Import from Parser (JSON) ready
- Structural SHA-256 hashing

# Changes — Diff Intelligence

Detect, classify, and audit every structural change

The screenshot displays the 'Changes' section of the Teradata Project SCION interface. It shows a comparison of snapshots #1 and #2, resulting in 19 total changes. The changes are categorized by severity: 13 Breaking, 7 High, 8 Medium, and 4 Low. A table lists the changes, including ID, Snapshot, Object Type, Object, Change, Severity, and Breaking status. The table shows changes to columns, tables, and table types, with details on the 'Before' and 'After' states for selected items.

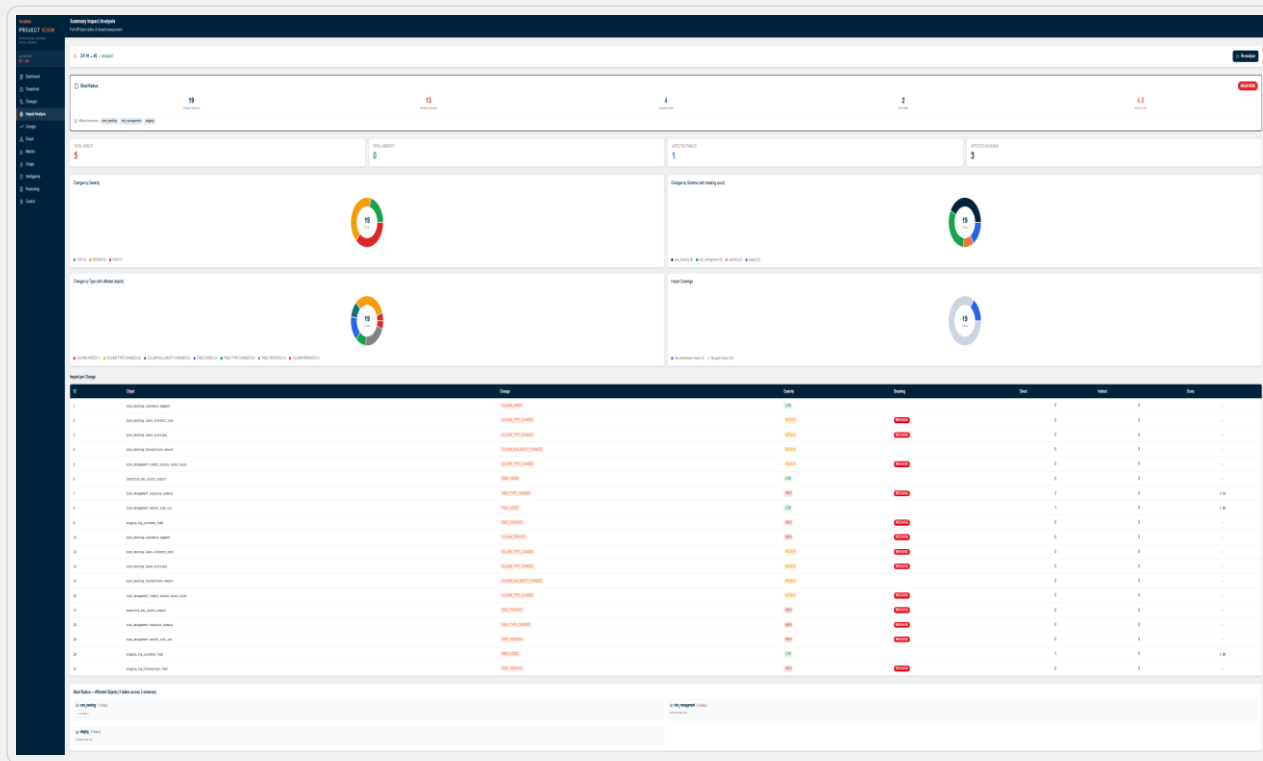
| ID | Snapshot | Object Type | Object                                   | Change                     | Severity | Breaking  |
|----|----------|-------------|------------------------------------------|----------------------------|----------|-----------|
| 1  | #1 → #2  | COLUMN      | 009_banking.customers.segment            | COLUMN_ADDED               | LOW      |           |
| 2  | #1 → #2  | COLUMN      | 009_banking.users.segment_date           | COLUMN_TYPE_CHANGED        | MEDIUM   | YES (NEW) |
| 3  | #1 → #2  | COLUMN      | 009_banking.users.segment_date           | COLUMN_TYPE_CHANGED        | MEDIUM   | YES (NEW) |
| 4  | #1 → #2  | COLUMN      | 009_banking.transactions.amount          | COLUMN_NULLABILITY_CHANGED | MEDIUM   | YES (NEW) |
| 5  | #1 → #2  | COLUMN      | 159_management.credit_scores.score_value | COLUMN_TYPE_CHANGED        | MEDIUM   | YES (NEW) |
| 6  | #1 → #2  | TABLE       | REPORTING.tbl_assets_report              | TABLE_ADDED                | LOW      |           |
| 7  | #1 → #2  | TABLE       | 159_management.expenses.summary          | TABLE_TYPE_CHANGED         | HIGH     | YES (NEW) |
| 8  | #1 → #2  | TABLE       | 159_management.request_loan_req          | TABLE_ADDED                | LOW      |           |
| 9  | #1 → #2  | TABLE       | staging.tbl_customers_feed               | TABLE_REMOVED              | HIGH     | YES (NEW) |
| 10 | #2 → #3  | COLUMN      | 009_banking.customers.segment            | COLUMN_REMOVED             | HIGH     | YES (NEW) |
| 11 | #2 → #3  | COLUMN      | 009_banking.users.segment_date           | COLUMN_TYPE_CHANGED        | MEDIUM   | YES (NEW) |
| 12 | #2 → #3  | COLUMN      | 009_banking.users.segment_date           | COLUMN_TYPE_CHANGED        | MEDIUM   | YES (NEW) |
| 13 | #2 → #3  | COLUMN      | 009_banking.transactions.amount          | COLUMN_NULLABILITY_CHANGED | MEDIUM   | YES (NEW) |
| 14 | #2 → #3  | COLUMN      | 159_management.credit_scores.score_value | COLUMN_TYPE_CHANGED        | MEDIUM   | YES (NEW) |
| 15 | #2 → #3  | TABLE       | REPORTING.tbl_assets_report              | TABLE_REMOVED              | HIGH     | YES (NEW) |
| 16 | #2 → #3  | TABLE       | 159_management.expenses.summary          | TABLE_TYPE_CHANGED         | HIGH     | YES (NEW) |
| 17 | #2 → #3  | TABLE       | 159_management.request_loan_req          | TABLE_REMOVED              | HIGH     | YES (NEW) |

## Key Features

- Cumulative diffs across snapshot ranges
- Severity: HIGH / MEDIUM / LOW
- Breaking vs Non-Breaking
- Before/after state with JSON evidence

# Impact Analysis — Blast Radius

Understand the full reach of every change

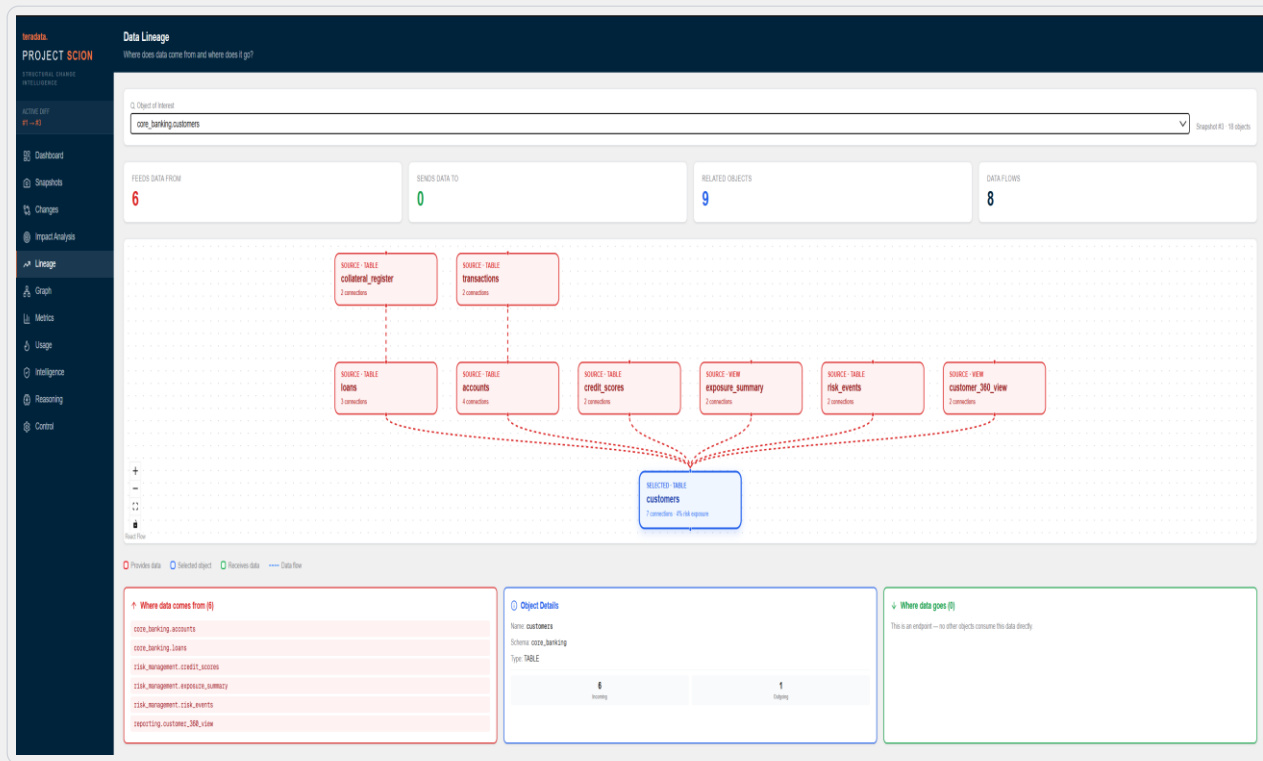


## Key Features

- Batch analysis across all changes
- Blast radius: schemas, tables, depth
- Impact score with depth decay
- Risk level classification

# Data Lineage — Visual Data Flow

Where does data come from and where does it go?



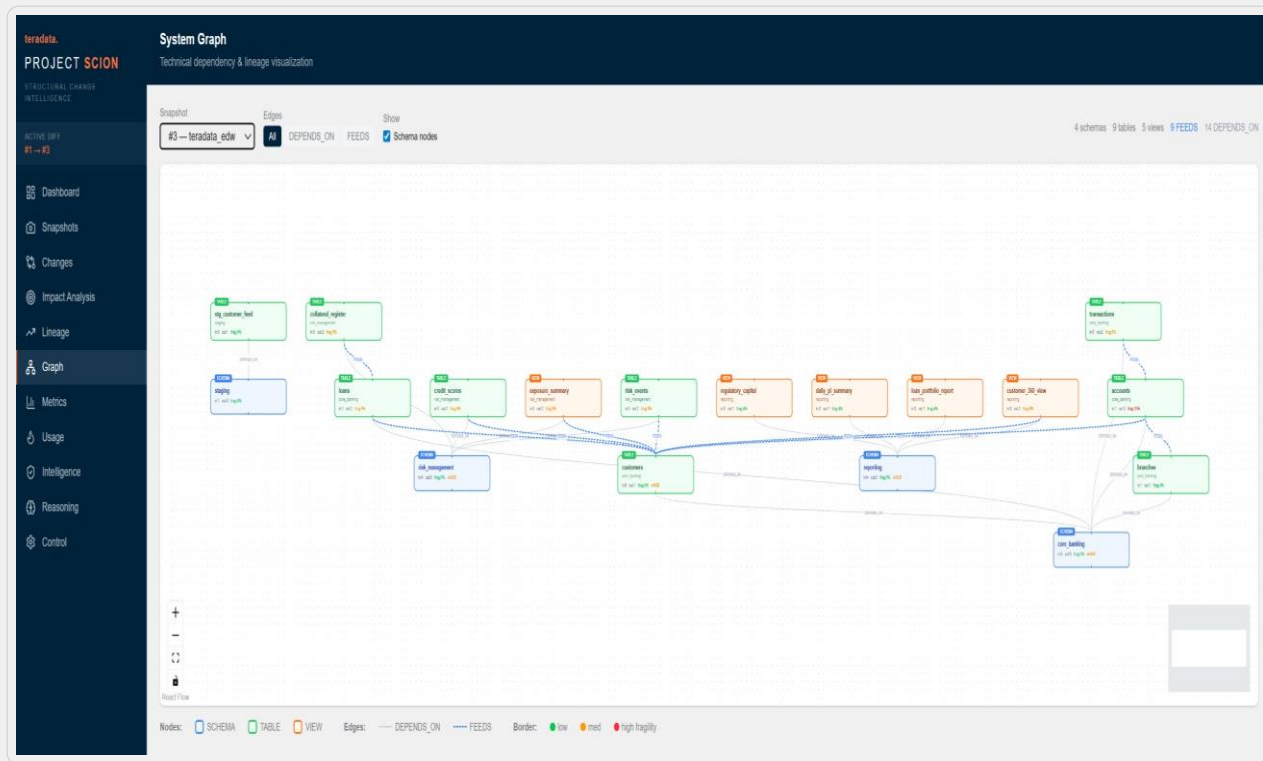
## Key Features

- Sources (red) → Selected (blue) → Consumers (green)
- Click to navigate between objects
- Connection metrics and risk exposure



# System Graph — Dependencies

Full structural dependency network

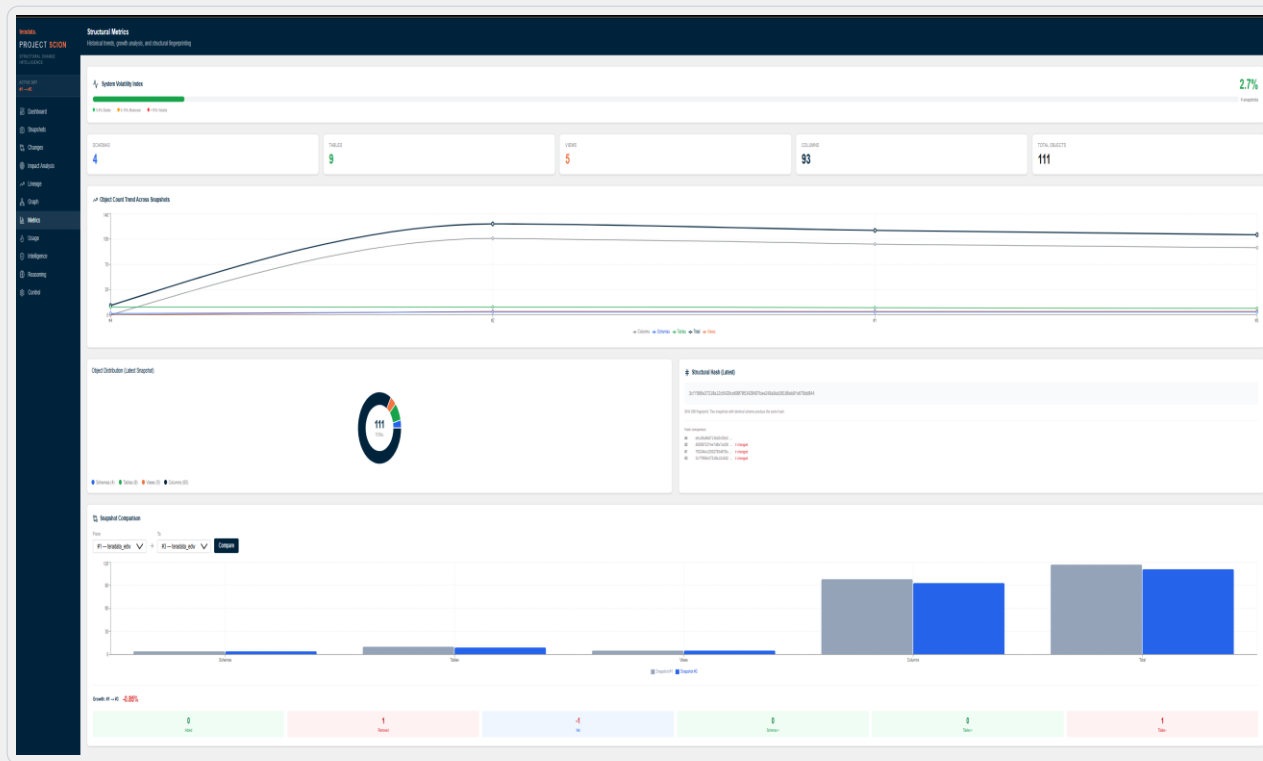


## Key Features

- Interactive visualization with zoom/pan
- FEEDS vs DEPENDS\_ON edges
- Fragility coloring & hub detection
- Filter by edge type

# Structural Metrics

Track how your EDW evolves over time

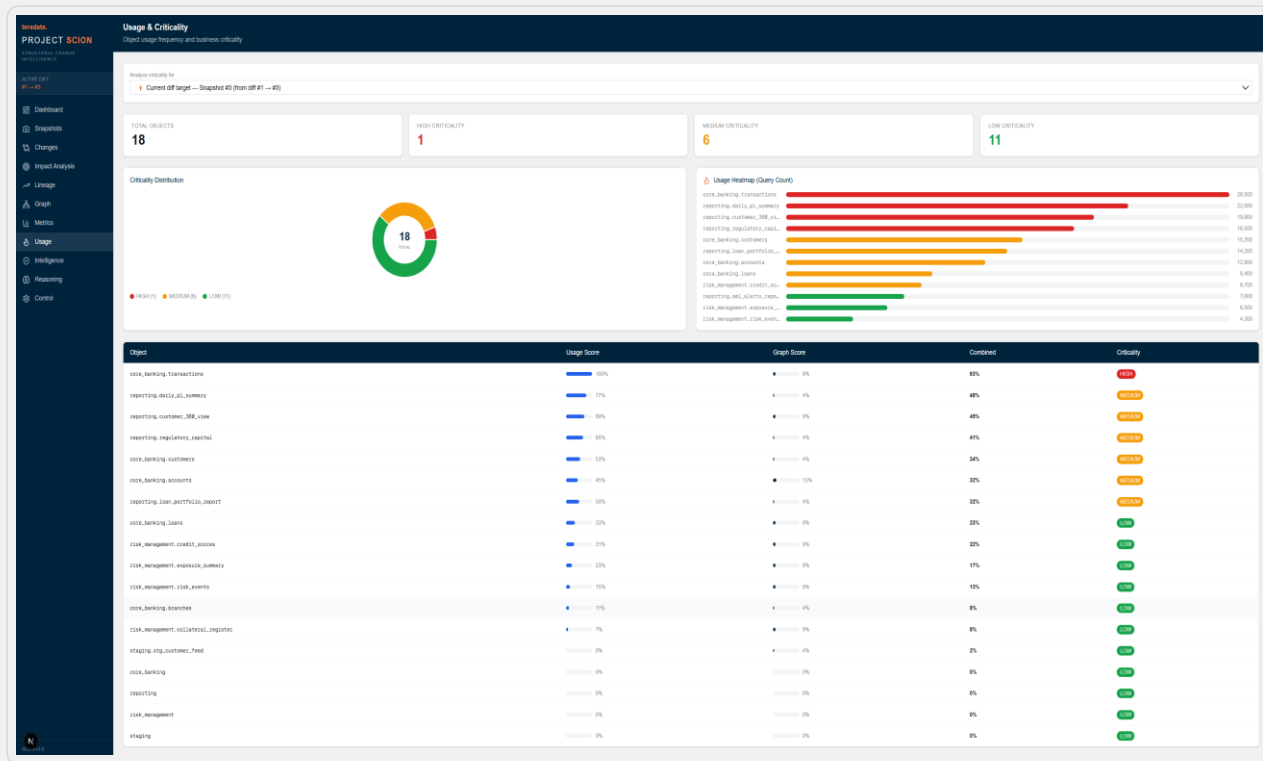


## Key Features

- System volatility index
- Object count trends across snapshots
- Side-by-side snapshot comparison
- Structural hash fingerprinting

# Usage & Criticality

Weight technical impact with real-world usage

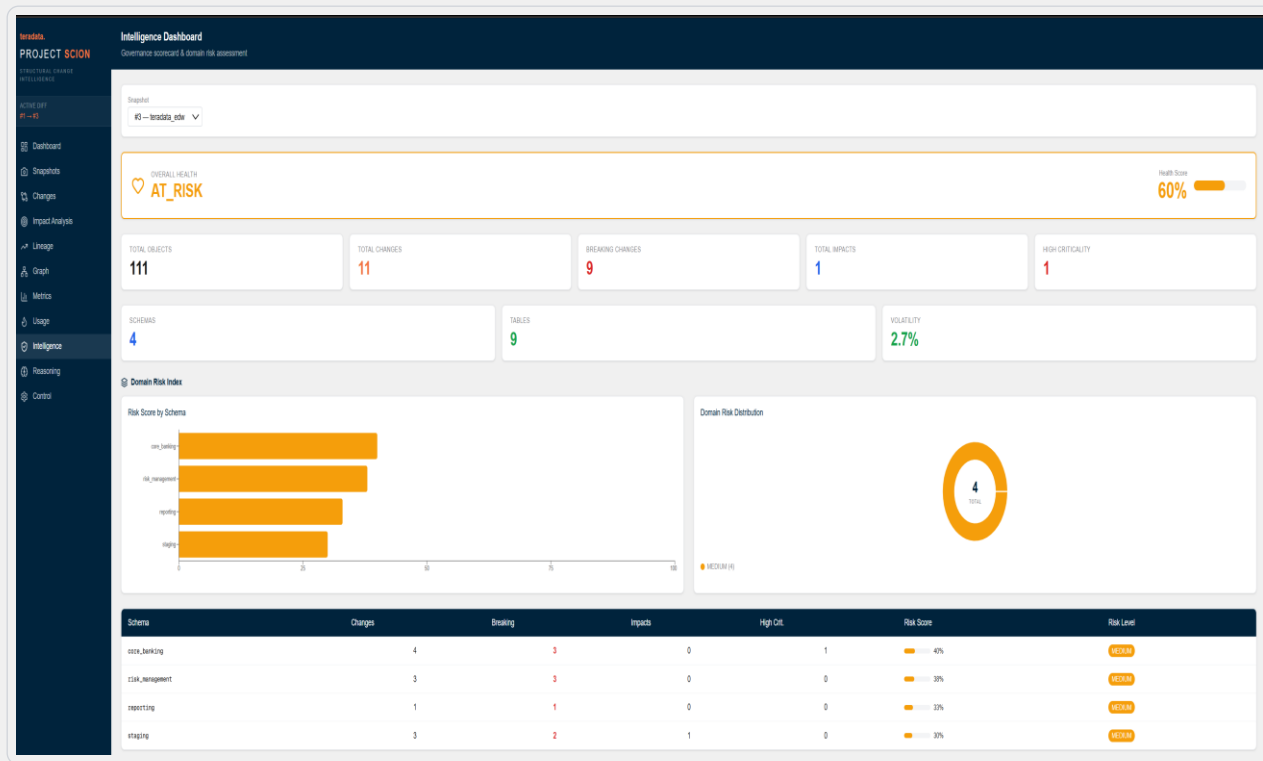


## Key Features

- Query log ingestion from parser
- Usage frequency heatmap
- Combined score (60% usage + 40% graph)
- HIGH/MEDIUM/LOW criticality

# Intelligence — Governance

Executive-ready structural health assessment

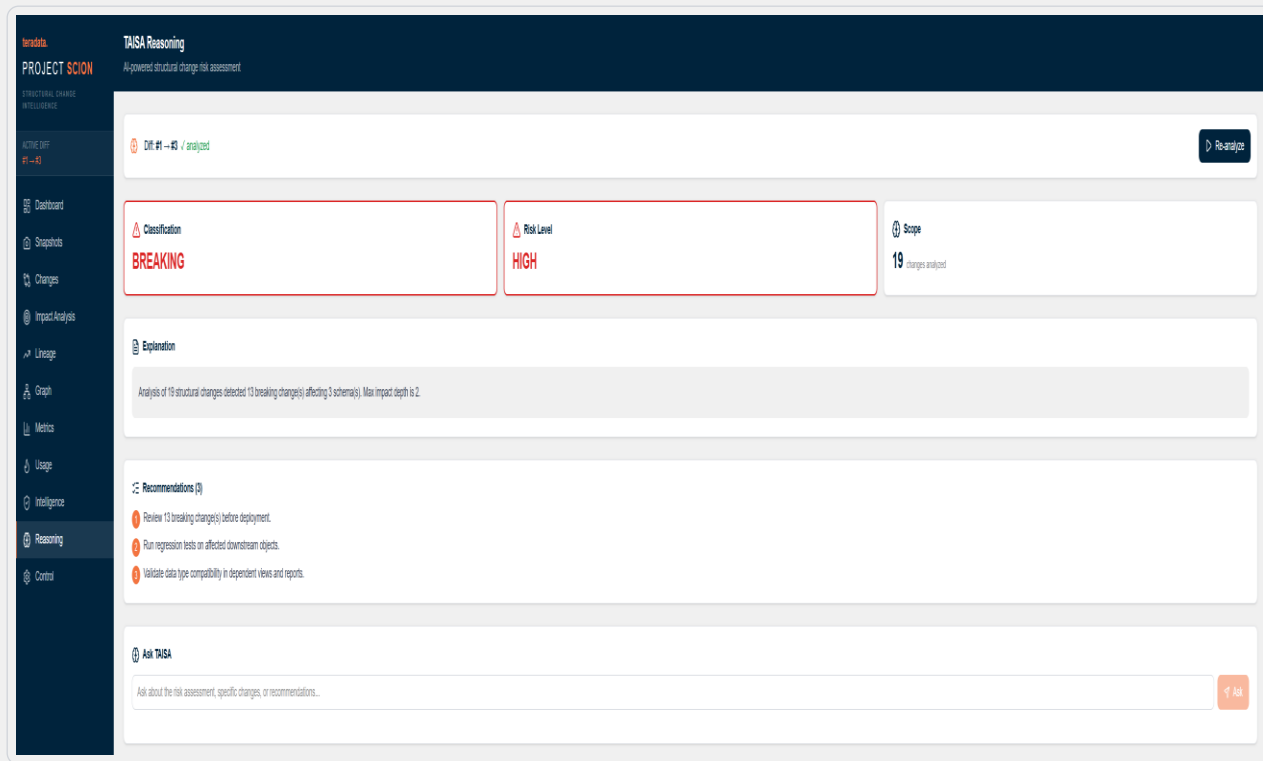


## Key Features

- Health score (Healthy / At Risk / Critical)
- Domain risk index per schema
- Breaking changes per domain
- Executive KPIs

# TAISA — AI Risk Assessment

LLM-powered analysis with recommendations



## Key Features

- BREAKING / NON\_BREAKING classification
- Risk: LOW / MEDIUM / HIGH / CRITICAL
- Actionable recommendations
- Interactive Q&A with AI

# SCION vs Kalido

Why build when you can buy? Because ownership beats rental.

|                  | Kalido                 | SCION                                |
|------------------|------------------------|--------------------------------------|
| Cost             | ~\$40K/month licensing | Zero — fully owned IP                |
| Transparency     | Black-box processing   | 100% auditable (JSON)                |
| AI               | None                   | LLM-powered via TAISA                |
| Customization    | Vendor-dependent       | Fully extensible                     |
| Change Detection | Opaque parsing         | Severity + breaking classification   |
| Impact Analysis  | Limited                | Blast radius + graph metrics         |
| Governance       | Basic                  | Scorecard + domain risk + volatility |

# Next Steps

From demo to production

1

**Connect real parser JSON**

Import button exists — map fields when format is final

2

**Connect to Teradata EDW**

TeradataAdapter exists — configure connection string

3

**Enable TAISA real mode**

One environment variable: SCION\_TAISA\_MODE=real

4

**Shadow-mode deployment**

Run alongside Kalido — compare results

5

**Production hardening**

PostgreSQL, Docker, security, performance tuning

**SCION is ready for demo and shadow-mode deployment**

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# PROJECT SCION

Structural Change Intelligence Platform

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Ready for deployment